1. Create an VM with inbound traffic allowed for SSH and HTTP on the Security group.
2. Allow only SSH inbound on the NACL.
3. Once the HTTP service is up on the VM.
4. Allow HTTP inbound on the NACL

Note:--

For http service, below are the commands

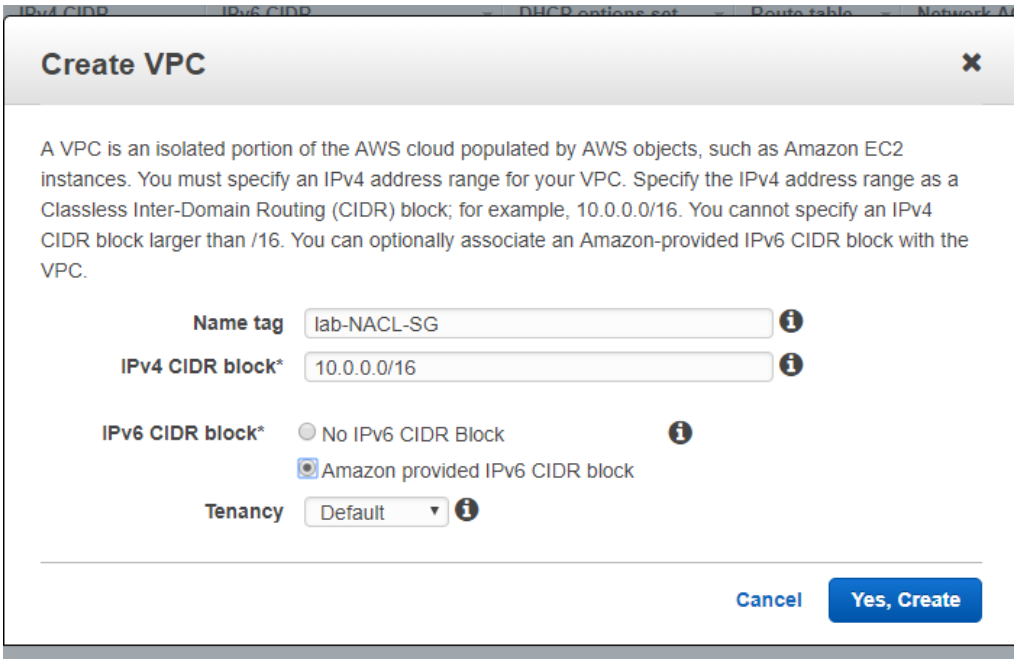
Sudo su

Yum install httpd

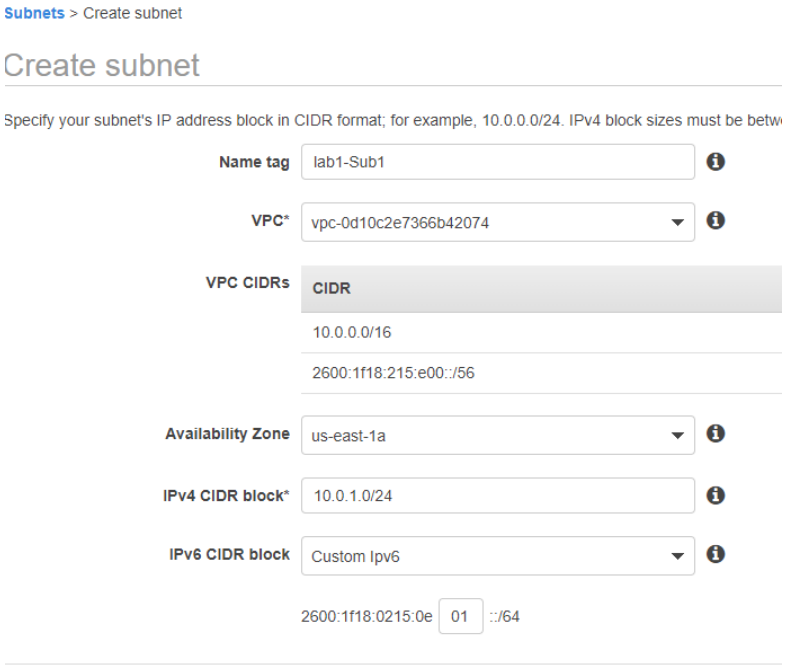
Service httpd start

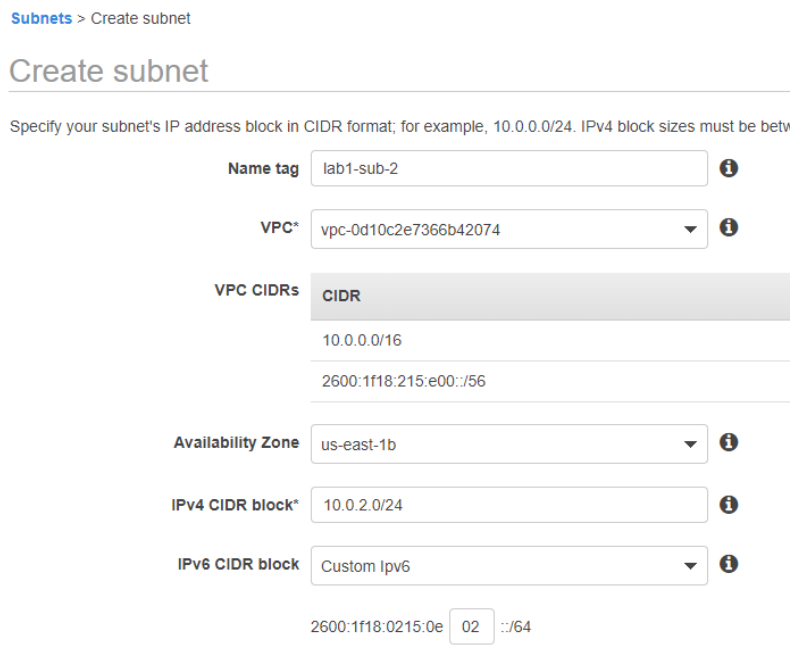
Solution:

1. **Create a VPC**



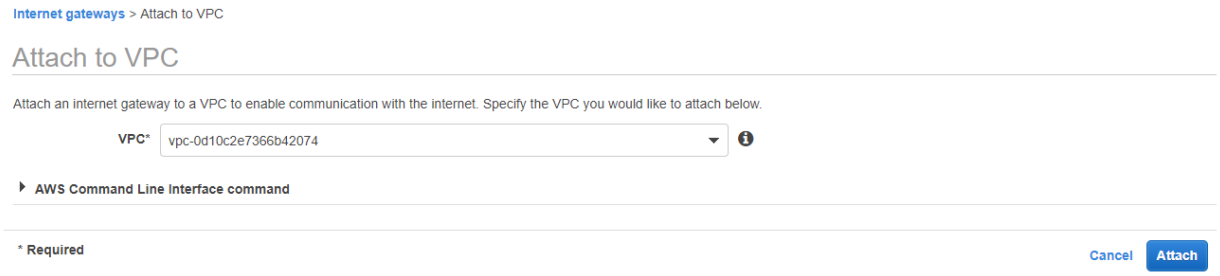
2. **Create 2 Subnets**





3. Create an Internet Gateway

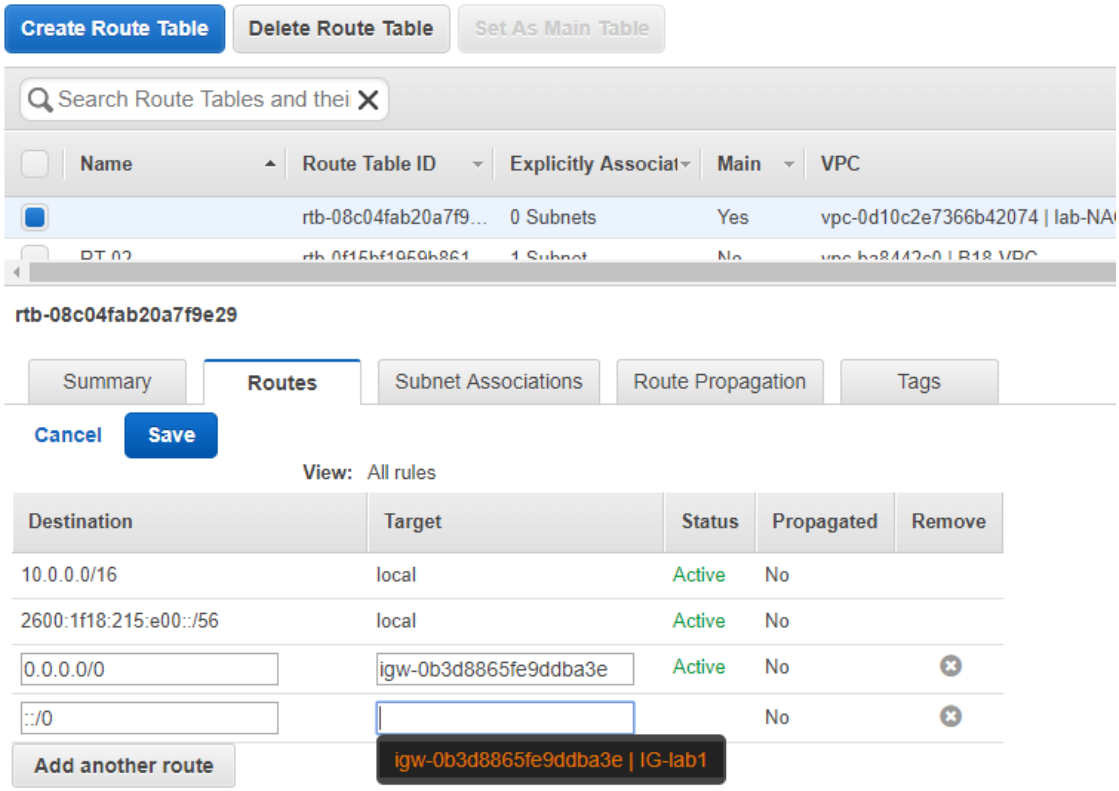
3.1 Attach the Internet Gateway to VPC



4. Update Routing Table –

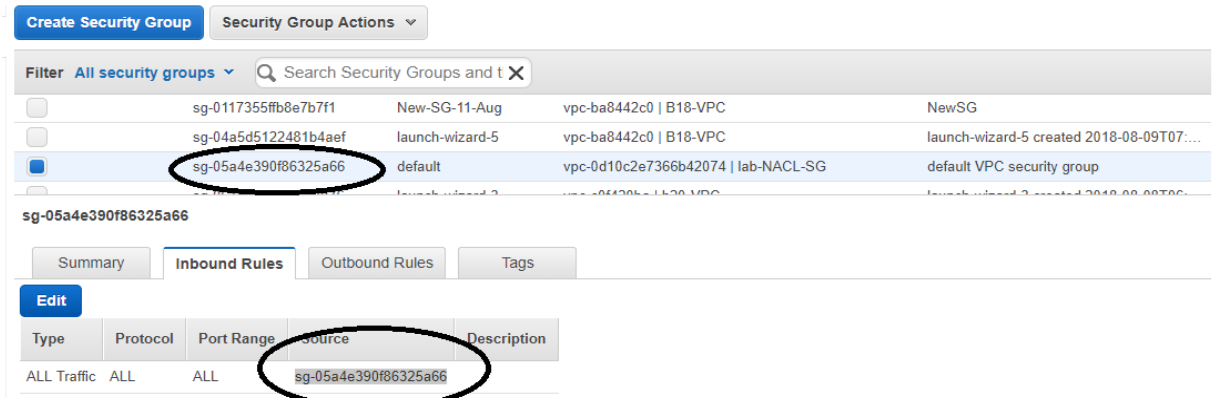
4.1 Add “0.0.0.0/0” to the default Routing Table for Ipv4.

4.2 Add “::/0” to the default Routing Table for Ipv6.

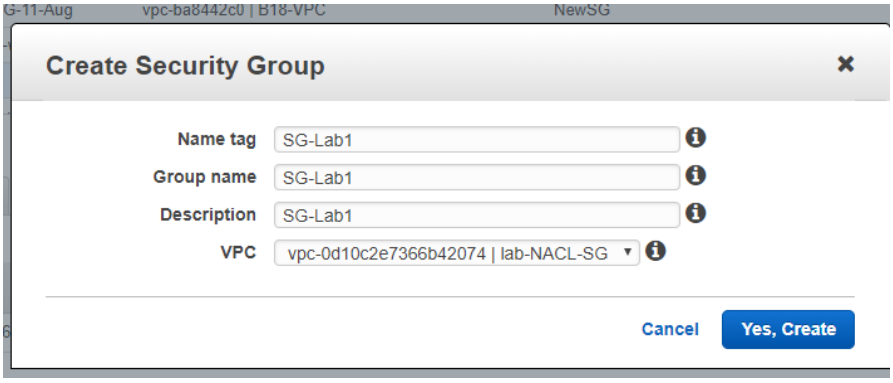


5. Create Security Group.

5.1 DONOT USE THE DEFAULT SECURITY GROUP.
Default Security Group is, OUTBOUND all Allow
And Inbound allow only itself, means Block ALL inbound.

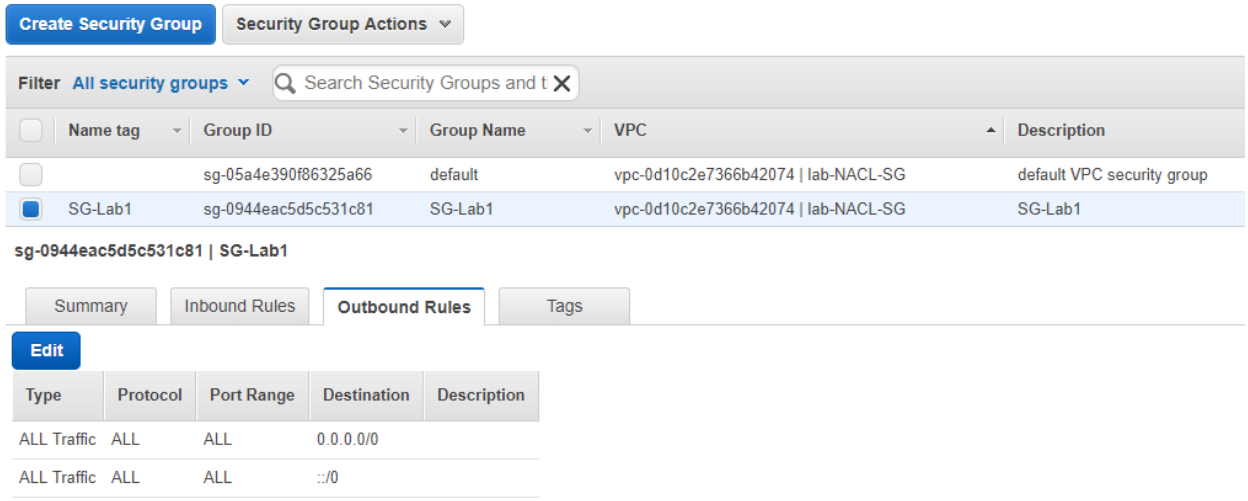


5.2 Create a new Security Group

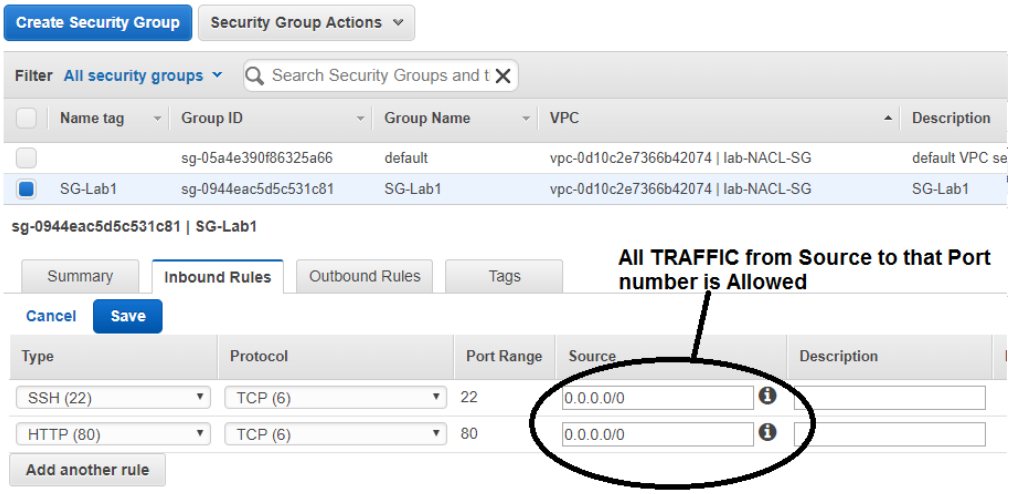


5.3 Add rules accordingly.

By default Outbound ALL TRAFFIC is allowed



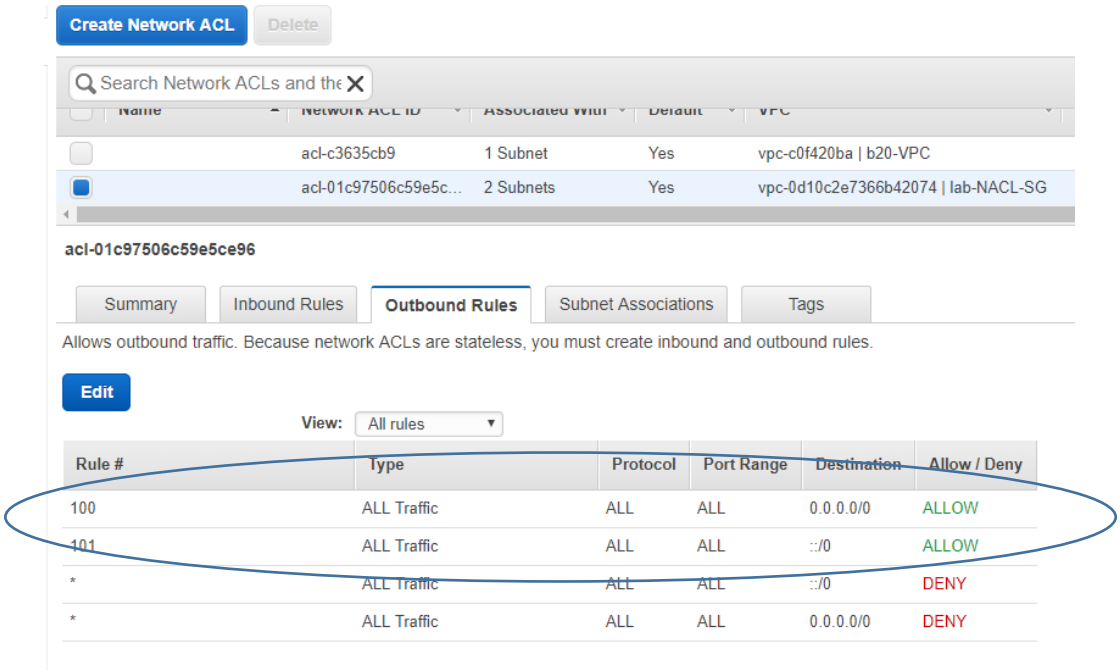
Inbound, allow only SSH and HTTP.

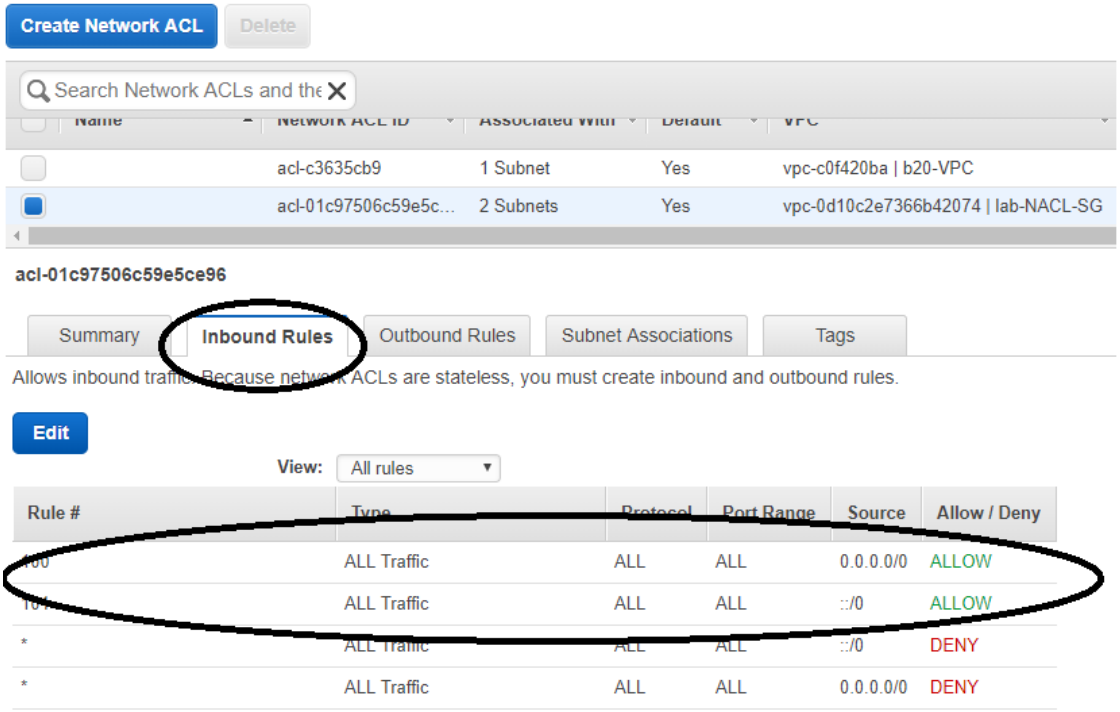


6. Create Network Access Control List.

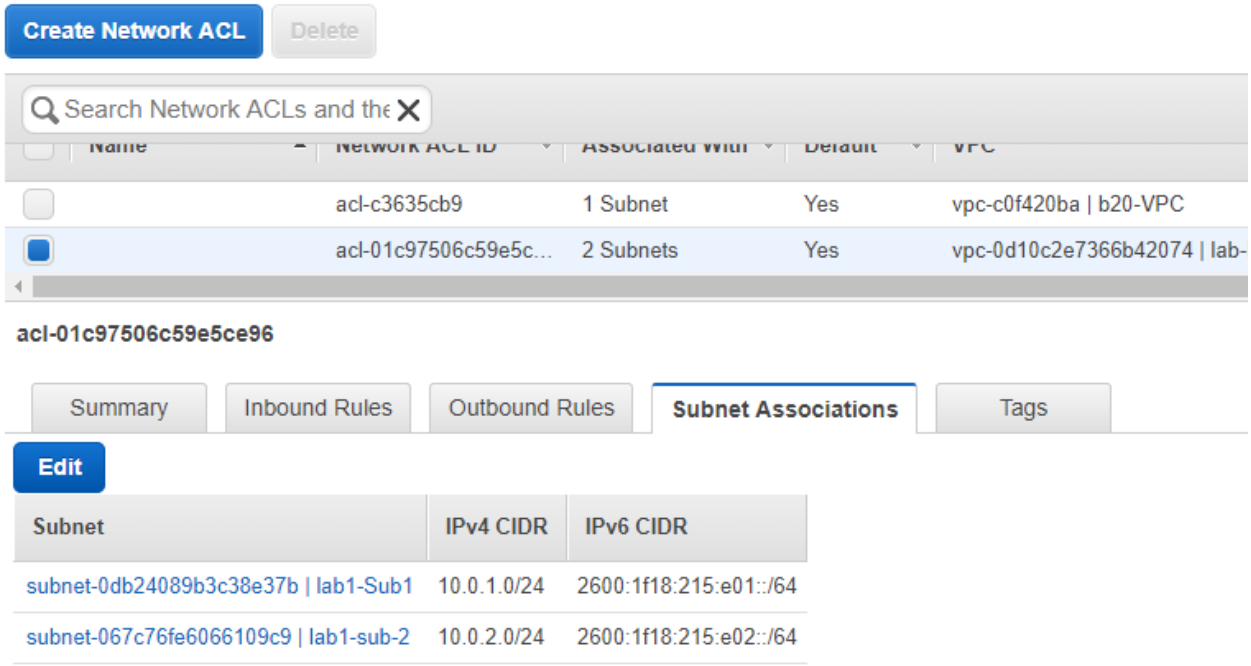
We WOULD be using the DEFAULT NACL created by VPC.

6.1 By default, ALL outbound and Inbound Traffic is Allowed

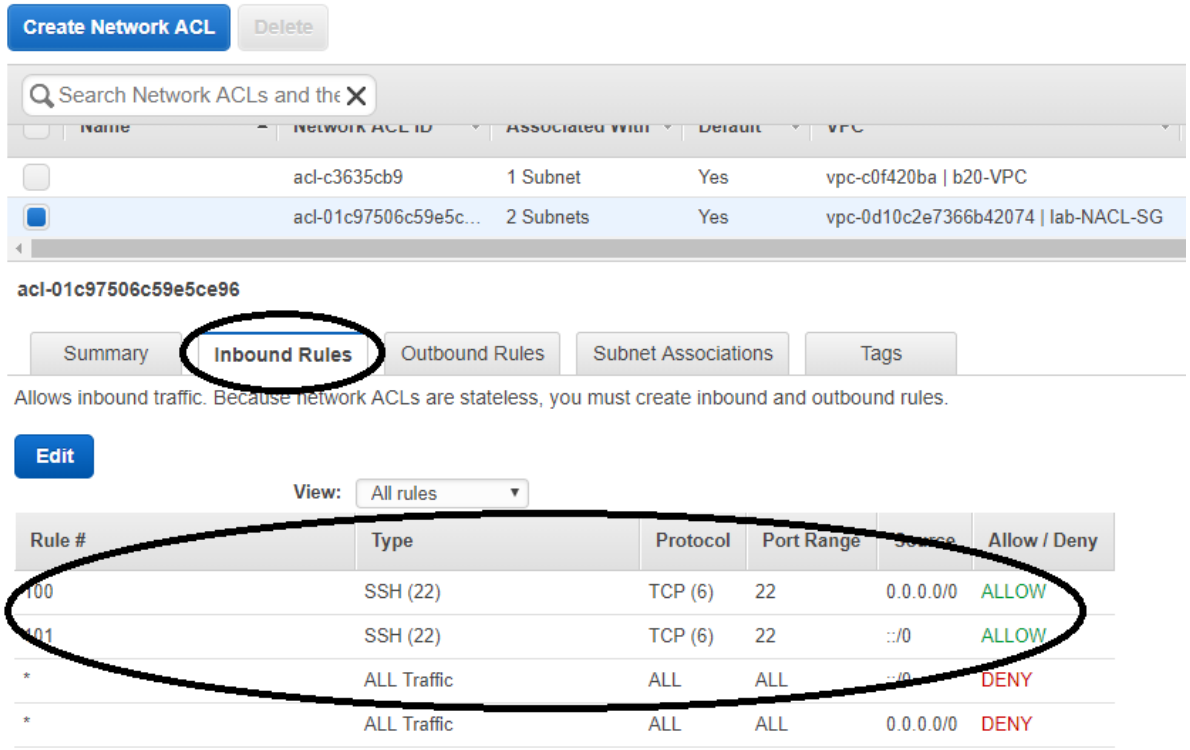




6.2 All the subnets are ASSOCIATED to the DEFAULT NACL



6.3 Add the NACL rules to the INBOUND as mentioned. – Allow only SSH.



7. **Create an EC2 instance in Subnet -1 with Public IPV6 and IPv4.**

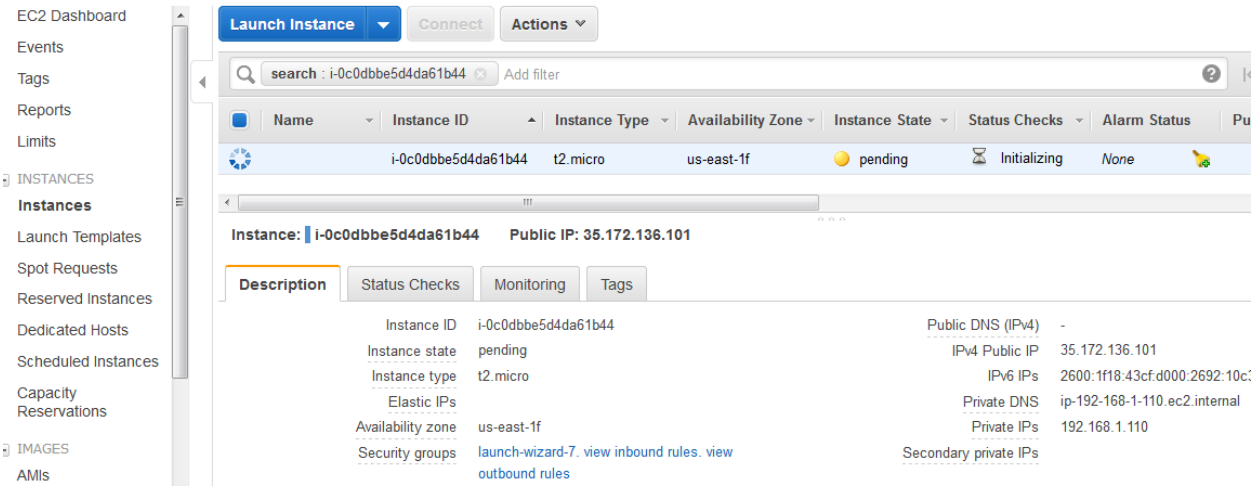
From CLI.

aws ec2 run-instances --image-id ami-b70554c8 --count 1 --instance-type t2.micro --key-name b16-key1 --security-group-ids sg-0944eac5d5c531c81 --subnet-id subnet-0db24089b3c38e37b

```
[c:\~]$ aws ec2 run-instances --image-id ami-b70554c8 --count 1 --instance-type t2.micro --key-name b16-key1 --security-group-ids sg-0944eac5d5c531c81 --subnet-id su
bnet-0db24089b3c38e37b

-----------------------------------------------------------------
|                          RunInstances                         |
+-------------------------+-------------------------------------+
|  OwnerId                |  766476266373                       |
|  ReservationId          |  r-05e98e8c7e56c8485                |
+-------------------------+-------------------------------------+
||                         Instances                           ||
|+-------------------------+-------------------------------+|
||  AmiLaunchIndex         |  0                            ||
||  Architecture           |  x86_64                       ||
||  ClientToken            |                               ||
||  EbsOptimized           |  False                        ||
||  Hypervisor             |  xen                          ||
||  ImageId                |  ami-b70554c8                 ||
||  InstanceId             |  i-01764bb24d4b2211d          ||
||  InstanceType           |  t2.micro                     ||
||  KeyName                |  b16-key1                     ||
||  LaunchTime             |  2018-08-11T09:04:51.000Z     ||
||  PrivateDnsName         |  ip-10-0-1-115.ec2.internal   ||
||  PrivateIpAddress       |  10.0.1.115                   ||
||  PublicDnsName          |                               ||
||  RootDeviceName         |  /dev/xvda                    ||
||  RootDeviceType         |  ebs                          ||
||  SourceDestCheck        |  True                         ||
||  StateTransitionReason  |                               ||
||  SubnetId               |  subnet-0db24089b3c38e37b     ||
||  VirtualizationType     |  hvm                          ||
||  VpcId                  |  vpc-0d10c2e7366b42074        ||
|+-------------------------+-------------------------------+|
|||                        Monitoring                       |||
||+-----------------------+-------------------------------+||
```

Or Create the EC2 instance from the console of AWS.



7.1 After creation of the EC2, login and try the below command

 # sudo su

 # yum install httpd -y

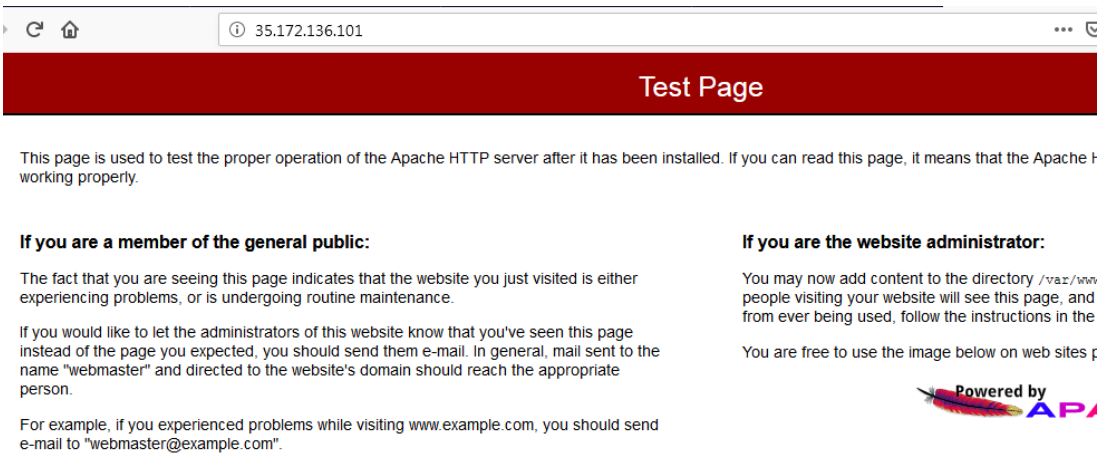
This would FAIL as “inbound” http is BLOCKED on NACL.

```
Total download size: 1.5 M
Installed size: 4.4 M
Downloading packages:
httpd-2.4.39-1.amzn2.0.1.x86_6 FAILED
http://amazonlinux.us-east-1.amazonaws.com/2/core/2.0/x86_64/11be506f86779fba9940dfce8be18d996726d1db22
585c19e638db48/../../../../../blobstore/728fd44d5948fae7430e2e6edff283b9cd5acb102e26bc00b90fa2cde504823
2.4.39-1.amzn2.0.1.x86_64.rpm?instance_id=i-0c0dbbe5d4da61b44&region=us-east-1: [Errno 14] curl#7 - "Fa
connect to amazonlinux.us-east-1.amazonaws.com port 80: Connection timed out"
Trying other mirror.
mod_http2-1.14.1-1.amzn2.x86_6 FAILED
http://amazonlinux.us-east-1.amazonaws.com/2/core/2.0/x86_64/11be506f86779fba9940dfce8be18d996726d1db22
585c19e638db48/../../../../../blobstore/498fc7b3a02385a5ce521a7b6dfc95fac165929e6c947841b8b2d265a19f7cb
tp2-1.14.1-1.amzn2.x86_64.rpm?instance_id=i-0c0dbbe5d4da61b44&region=us-east-1: [Errno 14] curl#7 - "Fa
connect to amazonlinux.us-east-1.amazonaws.com port 80: Connection timed out"
Trying other mirror.
```

All ALLOW TRAFFIC INBOUND for installation on NACL.

Then install the httpd.

 # yum install httpd
 # service httpd start.

Now again in the NACL , allow only SSH and HTTP.

You should be able to access the “http://<public ip on ec2> page.